

Reliability-based design optimization of wind turbine drivetrain with integrated multibody gear dynamics simulation considering wind load uncertainty

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Abstract This study aims to develop an integrated computational framework for the reliability-based design optimization (RBDO) of wind turbine drivetrains to assure the target reliability under wind load and gear manufacturing uncertainties. Gears in wind turbine drivetrains are subjected to severe cyclic loading due to highly variable wind loads that are stochastic in nature. Thus, the failure rate of drivetrain systems is reported to be higher than the other wind turbine components, and improving drivetrain reliability is critically important in reducing downtime caused by gear failures. In the numerical procedure developed in this study, a wide spatiotemporal variability for wind loads is considered using 249 sets of wind data to evaluate probabilistic contact fatigue life in the sampling-based RBDO. To account for wind load uncertainty in evaluation of the tooth contact fatigue, multiple drivetrain dynamics simulations need to be run under various wind load scenarios in the RBDO process. For this reason, a numerical procedure based on the multivariable tabular contact search algorithm is applied to the modeling of wind turbine drivetrains to reduce the overall computational time while retaining the precise contact geometry required for considering the gear tooth profile optimization. An integrated computational framework for the wind turbine drivetrain RBDO is then developed by incorporating the wind load uncertainty, the rotor blade aerodynamics model, the drivetrain dynamics model, and the probabilistic contact fatigue failure model. It is demonstrated that the

RBDO optimum for a 750 kW wind turbine drivetrain obtained using the procedure developed in this study can achieve the target 97.725% reliability (2 sigma quality level) with only a 1.4% increase in the total weight from the baseline design, which had a reliability of 8.3%. Furthermore, it is shown that the tooth profile optimization, tip relief introduced as a design variable, prevents a large increase of the face width that would result in a large increase in the weight (cost) of the drivetrain in order to satisfy the target reliability against the tooth contact fatigue failure.

Keywords Reliability-based design optimization · Wind turbine drivetrain · Wind load uncertainty · Multibody dynamics · Gears · Contact fatigue life · Reliability

1 Introduction

Gear teeth of a wind turbine drivetrain are subjected to severe cyclic rolling contact resulting from highly variable wind loads, which are stochastic in nature. Thus, contact fatigue is one of the leading causes of unintended gearbox failure that can prevent wind turbines from achieving the expected service life. Furthermore, it has been reported that the failure rate of gearboxes is higher than other wind turbine components (Sheng et al. 2011; Errichello and Muller 2012). Improvement of the drivetrain reliability is, therefore, critically important in reducing downtime caused by gear failures of wind turbine systems. Since wind loads are stochastic and highly variable over the period of service life, use of deterministic contact fatigue models for wind turbine drivetrains can lead to unrealistic gearbox product life prediction. In addition, gear design is a complicated iterative process, involving many design variables, requirements, and constraints (Vanderplaats et al. 1988). Most existing gear design optimizations have been carried out with an

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assumption of deterministic load conditions to find an optimum contact ratio, face width, and tooth tip and root profiles (Spitas and Spitas 2007; Sansalvador and Jauregui 1993; Kapelevich and Shekhtman 2009). Minimization of gear transmission error, which is the major source of gearbox noise and vibration, can be achieved by modifying the gear tooth profile through the lead crowning and tip/root relief (Maatar and Velez 1997; Velez et al. 2011; Sundaresan et al. 1991). Since the tooth profile modification is on an order of microns and it is in the same order of manufacturing variance, a robust design optimization using Taguchi Method is adopted in the literature (Sundaresan et al. 1991) to make the gear profile design insensitive to the manufacturing variance. It is also shown that the maximum contact pressure due to cyclic contact loads can be lowered by tooth profile modification (i.e., tip relief) (Osman and Velez 2011), thereby lowering the maximum shear stress beneath the contact surface, which is a major driving factor for crack growth of rolling contact fatigue. To account for the effect of profile modification on the contact fatigue life, a gear contact model has to be capable of predicting the precise gear tooth contact geometry (i.e., microgeometry) as well as the mesh stiffness variation.

To account for wind load uncertainty, a probabilistic fatigue life prediction for a wind turbine drivetrain is addressed in recent literature (Dong et al. 2013; Alemayehu and Osire 2015), in which wind load uncertainty is defined solely by the probability density function (PDF) of the mean wind speed using the generalized gamma function and then contact fatigue life is evaluated using the resulting random maximum contact pressure (Dong et al. 2013). Furthermore, to meet the target reliability against gear tooth contact failures, reliability-based design optimization (RBDO) has to be pursued for geared systems. RBDO has proven to be an effective approach that enables reliable and cost-effective design for a wide variety of engineering applications under uncertainties (Tu et al. 1999; Youn and Wang 2008; Rahman and Wei 2008; Lee et al. 2011) and RBDO has been applied to wind turbine rotor blades in the literature (Hu et al. 2016a) considering wide spatiotemporal wind load uncertainty.

On the other hand, despite many studies on the gear contact and failure analysis of wind turbine drivetrains (Dong et al. 2013; Alemayehu and Osire 2015; Oyague 2009; Helsen et al. 2011; Vanhollebeke et al. 2015), RBDO of wind turbine drivetrains has not yet been addressed considering wide spatiotemporal wind load uncertainty and variability of gear design variables. Furthermore, a wind turbine is a multi-physics system involving mechanical and aerodynamics simulations to predict the system response, thus an integrated simulation framework is required such that the wind load uncertainty, random wind field generation, rotor blade aerodynamics, drivetrain dynamics considering the precise gear tooth contact geometry, and contact fatigue models can be fully integrated into the design optimization framework. Therefore, this study is aimed to develop an integrated multibody dynamics

computational framework for RBDO of wind turbine drivetrains considering wide spatiotemporal wind load uncertainty to meet the target reliability for a 20-year service life. The main contribution of this paper lies in the following points:

1. An integrated multibody dynamics computational framework for RBDO of wind turbine drivetrains is developed using the multibody drivetrain dynamics simulation capability that incorporates wide spatiotemporal wind load uncertainty, probabilistic gear tooth contact fatigue model, and rotor blade aerodynamics model.
2. In order to reduce the computational cost of multibody gear dynamics simulations considering uncertainty associated with wind loads and design variables in the RBDO process, the multivariable tabular tooth contact search algorithm is applied to the wind turbine drivetrain dynamic analysis for evaluation of probabilistic contact fatigue damages. This leads to computationally efficient drivetrain dynamics simulation suited for RBDO while retaining the precise contact geometry associated with the tooth profile optimization (i.e., tip relief).
3. It is demonstrated that the RBDO optimum for a 750 kW wind turbine drivetrain obtained using the procedure developed in this study can achieve the target 97.725% reliability (2σ quality level) with only a 1.4% increase in the total weight from the baseline design, which has a reliability of 8.3%.
4. It is shown that the tooth profile optimization (tip relief) introduced as a design variable allows for preventing a large increase of the face width that would result in a large increase in the weight (cost) of the drivetrain for satisfying the target reliability against the tooth contact fatigue failure.

2 Wind load uncertainty

In order to account for wind load uncertainty for wind turbine systems, 10-min wind data is widely used to characterize the short-term wind load variability at a specific location (Veers and Winterstein 1998; Hu et al. 2016b). In this study, the joint probability density function (PDF) of the 10-min mean wind speed (V_{10}) and 10-min turbulence intensity (I_{10}) is introduced to characterize wind load uncertainty (Hu et al. 2016b). The turbulence intensity accounts for the severity of the wind speed fluctuation and is defined by $I_{10} = \Sigma_{10}/V_{10}$, where Σ_{10} is the standard deviation of the wind speed for the 10-min wind data. Due to this mathematical relation, the joint PDF of V_{10} and I_{10} is derived through the 10-min joint PDF of V_{10} and Σ_{10} using the copula density function $c_{V\Sigma}$ for wind data measured for one year as proposed in the literature (Hu et al. 2016b)

$$f_{VI}(v_{10}, i_{10}) = c_{V\Sigma}(v_{10}, \sigma_{10}) \cdot f_{V10}(v_{10}) \cdot f_{\Sigma10}(\sigma_{10}) \cdot v_{10} \quad (1)$$

where v_{10} , i_{10} and $\sigma_{10} = v_{10} \cdot i_{10}$ are realizations of random variables V_{10} , I_{10} and Σ_{10} , respectively. In the preceding equation, $f_{V_{10}}$ and $f_{\Sigma_{10}}$ are, respectively, the marginal PDFs for V_{10} and Σ_{10} . They are identified by the measured wind data using Weibull and gamma distributions as

$$f_{V_{10}}(v_{10}; C, k) = \frac{k}{C} \left(\frac{v_{10}}{C}\right)^{k-1} \exp\left[-\left(\frac{v_{10}}{C}\right)^k\right] \quad (2)$$

and

$$f_{\Sigma_{10}}(\sigma_{10}; a, b) = \frac{1}{b^a \Gamma(a)} \sigma_{10}^{a-1} \exp\left(-\frac{\sigma_{10}}{b}\right) \quad (3)$$

where C and k are the scale and shape parameters for the Weibull distribution for V_{10} , respectively. In the gamma distribution for Σ_{10} , a and b are the shape and scale parameters, respectively, and $\Gamma(a)$ is a gamma function of the parameter a . The Gumbel copula density function is given as (Hu et al. 2016b; Noh et al. 2009; Noh et al. 2010)

$$c_{V\Sigma}(u, v; \theta) = \frac{(-\ln u)^{\theta-1} (-\ln v)^{\theta-1} \left(\theta + w^{1/\theta} - 1 \right) w^{1/\theta-2} \exp\left(-w^{1/\theta}\right)}{uv} \quad (4)$$

where u and v are the marginal cumulative distribution function (CDF) values of V_{10} (Weibull distribution) and Σ_{10} (gamma distribution), respectively. w is given by $w = (-\ln u)^\theta + (-\ln v)^\theta$, where θ is defined by using Kendall's tau, as $\theta = 1/(1 - \tau)$. To define the joint PDF of V_{10} and I_{10} , the following five parameters need to be identified:

$$\mathbf{y} = [C \ k \ a \ b \ \tau]^T \quad (5)$$

It is important to notice here that one joint PDF characterizes wind load uncertainty measured at one location in a year. However, the wind load distribution varies at different locations (i.e., wind farms) in different years and such a wide spatiotemporal variability needs to be considered in the wind uncertainty model (Hu et al. 2016a). To this end, using 249 groups of wind data collected from different sites in different years, 249 sets of C , k , a , b , τ parameters are identified and each set of parameters represents the annual wind load variation at a specific location for a specific year. Using the maximum likelihood estimate method, the distribution types for the C , k , a , b , τ parameters are identified as the log-logistic, normal, generalized extreme value, Weibull, and extreme value distributions, respectively (Hu et al. 2016a). Accordingly, the 10-min joint PDF is defined as $f_{V,I}(\mathbf{y}; v_{10}, i_{10})$, where $\mathbf{y}$ is the realization of the PDF parameters given by Eq. (5). It is important to notice here that wind speeds are measured at a different height above ground for each set of data, thus the measured wind speeds need to be converted to those at the hub height of the wind turbine under consideration using the

normal wind profile model (International Organization for Standardization 2005).

With the 10-min mean wind speed v_{10} and turbulence intensity i_{10} , random time-domain wind speed data is generated using National Renewable Energy Laboratory (NREL) TurbSim software (Jonkman 2009). TurbSim can generate the spatial time-domain wind data using a prescribed power spectral density function (PSD) and a coherence function using Veer's method (Jonkman 2009). The PSD function is given as a function of both v_{10} and i_{10} , while the coherence function is given as a function of v_{10} only. The Veer's method is based on a general random process to generate time series of wind speed at spatial discrete points as $\mathbf{U}^{ij}(v_{10}^i, i_{10}^j) = [u^{ij}(t) \ v^{ij}(t) \ w^{ij}(t)]^T$. The wind speed data is then inputted into NREL FAST software (Jonkman and Buhl 2005) to perform the time-domain coupled nonlinear aero-hydro-servo-elastic simulation of a wind turbine considering the pitch control of rotor blades. The wind turbine blade pitch control in Region 3 is considered in this study to maintain a rated rotor speed (Jonkman and Buhl 2005). The resulting 10-min speed of the rotor hub is used as an input to the high-fidelity multibody drivetrain dynamics simulation for contact fatigue evaluation under various wind scenarios considered in the wind load uncertainty model using Monte Carlo simulation (MCS). This numerical procedure is schematically summarized in Fig. 1.

3 Wind turbine drivetrain dynamics model

Since multiple 10-min drivetrain dynamics simulations need to be run for various v_{10} and i_{10} values to account for the wide spatiotemporal wind load variability as shown in Fig. 1, the time-domain multibody drivetrain dynamics simulation can become a computational burden in the entire RBDO process. Furthermore, the tooth contact pressure is sensitive to the gear tooth profile geometry, thus an accurate description of the tooth profile geometry and precise contact geometry calculation are of crucial importance in accurate prediction of the contact fatigue life. Thus, use of a simplified lumped-mass gear dynamics model is not suited in this study. For example, profile modification called a tip relief is widely used to lower the transmission error that causes gear noises and vibrations (Velex et al. 2011; Sundaresan et al. 1991; Osman and Velex 2011). Such a slight modification of the tooth profile geometry in the order of microns alters the contact pressure on the profile surface, and it influences the contact fatigue damage (i.e., crack growth) as well since the crack growth is driven by the maximum shear stress beneath the surface. For this reason, the amount of tip relief is one of the important design variables for gears (Velex et al. 2011; Sundaresan et al. 1991; Osman and Velex 2011) and, in order to account for the effect of the tip

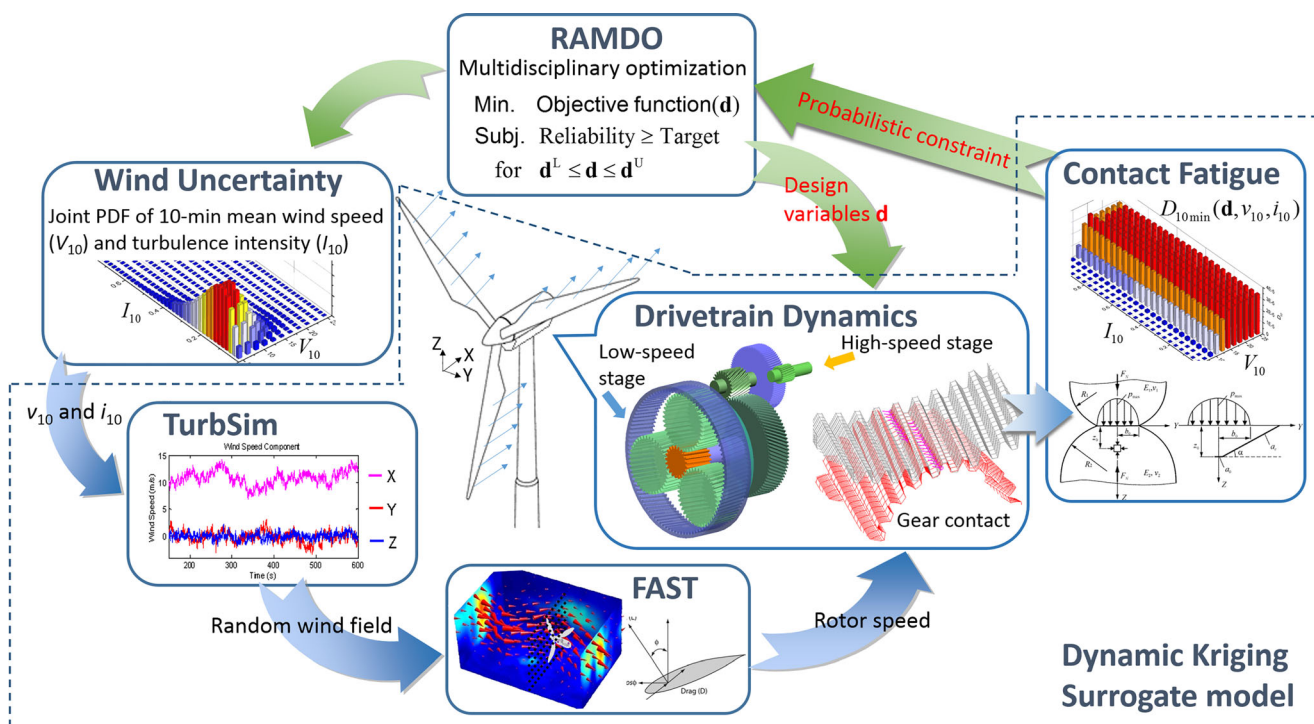
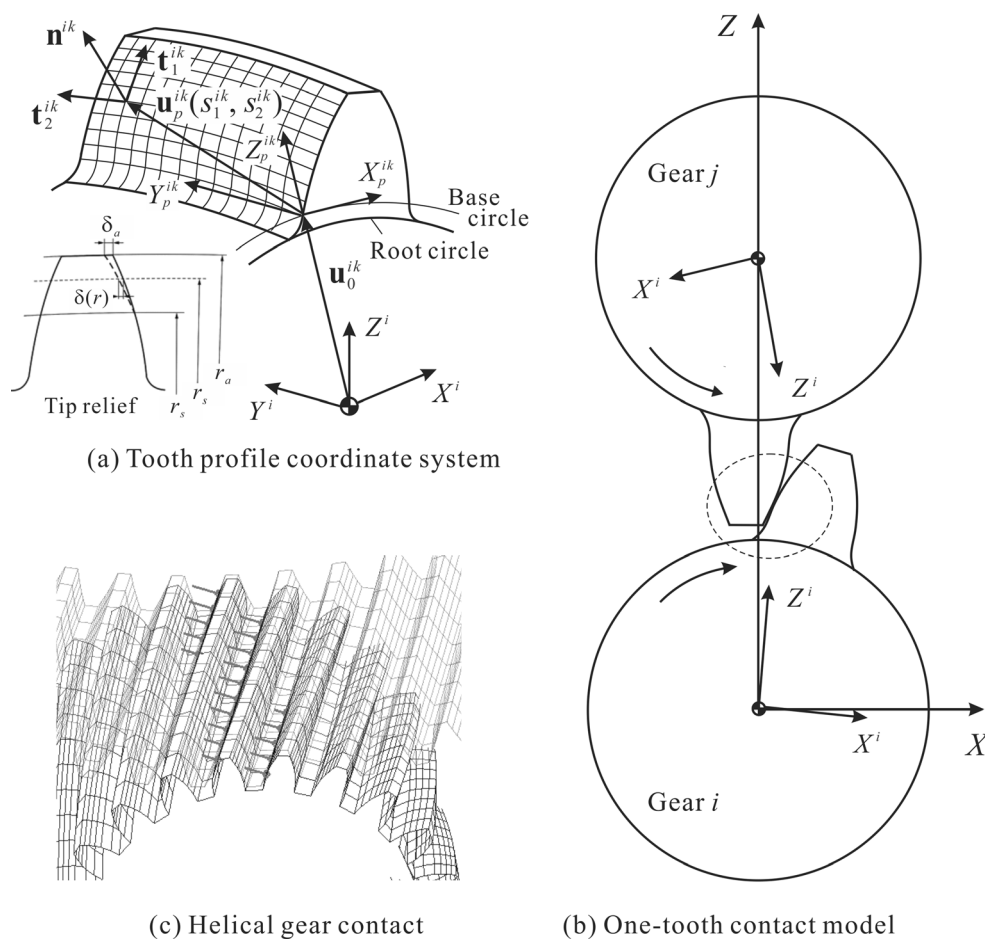


Fig. 1 Overview of integrated multibody gear dynamics simulation for wind turbine drivetrain RBDO considering wind load uncertainty

Fig. 2 Gear coordinate systems



relief, a microgeometry effect needs to be precisely considered in the gear dynamics simulation model.

3.1 Gear dynamics model with tabular contact search algorithm

The global position vector of a contact point on tooth k of rigid gear body i shown in Fig. 2a is written as

$$\mathbf{r}^{ik} = \mathbf{R}^i + \mathbf{A}^i \bar{\mathbf{u}}^{ik} \quad (6)$$

where $\mathbf{R}^i$ is the global position vector of the origin of the body coordinate system attached to the center of gear body; $\mathbf{A}^i$ is the rotation matrix; and $\bar{\mathbf{u}}^{ik}$ defines the local position vector of the contact point defined with respect to the body coordinate system. The gear tooth geometry is parameterized by the two surface parameters s_1^{ik} and s_2^{ik} , and the local position vector $\bar{\mathbf{u}}^{ik}$ is expressed as (Li et al. 2015)

$$\bar{\mathbf{u}}^{ik}(s_1^{ik}, s_2^{ik}) = \bar{\mathbf{u}}_0^{ik} + \mathbf{A}_0^{ik} \bar{\mathbf{u}}_p^{ik}(s_1^{ik}, s_2^{ik}) \quad (7)$$

where $\bar{\mathbf{u}}_0^{ik}$ and $\mathbf{A}_0^{ik}$ define the location and orientation of the tooth profile coordinate system with respect to the body coordinate system, respectively. The location of the contact point on the tooth profile is defined by $\bar{\mathbf{u}}_p^{ik}(s_1^{ik}, s_2^{ik})$ with respect to the profile coordinate system using the surface parameters in either analytical or numerical form. The linear tip relief as shown in Fig. 2a can also be considered in the tooth profile to reduce the gear transmission error at an operating torque. The amount of the tip relief is defined by

$$\delta(r) = \delta_a \frac{r - r_s}{r_a - r_s} \quad (8)$$

where r_s is the point that the tip relief starts and its end point is defined by r_a . The amount of tip relief at the end point is defined by δ_a . In the tooth profile optimization, the tooth profile geometry modification leads to a change in the contact pressure distribution over the contact surface, thus consideration of the precise micro contact geometry is of crucial importance in successful gear tooth profile optimization. This, however, leads to a computationally intensive procedure due to a need for iterative online collision detection for all pairs of gears in the complex multi-stage drivetrain model. Furthermore, for each design candidate, numerous simulations are required to account for wind load uncertainty as well as the variability of design variables in sampling-based RBDO. For this reason, a tabular contact search algorithm based on the combined nodal and non-conformal contact search algorithm proposed in the literature (Li et al. 2015; Sugiyama et al. 2012) is introduced and applied to the modeling of wind turbine drivetrains for RBDO.

Since the gear tooth contact is periodic, the solution to the contact geometry problem of one-tooth contact model as shown in Fig. 2b can be repeatedly used for detecting the contact of all teeth for a single pair of gears. For this reason, in this approach, the contact geometry analysis for a single tooth model is performed prior to the dynamic simulation to find the location of the contact point for one mesh cycle and then the location of the contact point on the tooth surface at various angles of the gear are stored in the multivariable look-up contact table. The multivariable look-up contact tables for all pairs of gears in a drivetrain model are then interpolated to locate contact points and their geometric properties used for the Hertzian contact calculation at every time step in the dynamics simulation. Due to the use of pre-calculated contact geometry for a pair of gears, the numerical procedure is computationally efficient while retaining the precise gear contact geometry for any modified tooth profiles (Sugiyama et al. 2012). Details on the interpolation procedure as well as the contact geometry analysis using the look-up table coordinate systems are provided in the literature (Li et al. 2015) for a complex planetary gear.

Using this approach, the helical gear tooth profile can also be considered by discretizing the gear tooth surface into a number of slices across the face width as shown in Fig. 2c (Flodin and Anderson 2001). Since the tooth surface is cut into a number of slices, tabular contact search can be utilized for all the sliced profiles using the same look-up contact table. It is, however, important to notice here that the look-up table coordinate system defined on the sliced profile needs to be rotated by $\Delta\theta$ from one sliced profile to the other to account for the helix curve. $\Delta\theta$ is given by (Litvin and Fuentes 2004)

$$\Delta\theta = \frac{B \cdot \cos\alpha_t \cdot \tan\beta_h}{n_s \cdot r_b} \quad (9)$$

where B is the tooth face width, α_t is the transverse pressure angle, β_h is the helix angle, r_b is the base radius, and n_s is the number of slices. The load distribution across the face width of helical gears is then described as a collection of forces acting on the sliced tooth profiles. The gear mesh force is defined by a penalty approach and Coulomb friction model is assumed for the tangential force. The mesh stiffness accounts for the effect of the contact stiffness k_{N1} (Tavakoli and Houser 1986), the tooth bending stiffness k_{N2} (Tavakoli and Houser 1986), and the gear body (foundation) stiffness k_{N3} (Sainsot and Velex 2004). Accordingly, the resulting equivalent mesh stiffness varies from the bottom to the tip of the gear tooth to capture the mesh force variation and an effect of gear tooth bending stiffness as well as gear body stiffness is considered in the rigid-body gear dynamics model through the mesh stiffness for calculating normal contact forces. More details on the mesh stiffness calculation and

validation are found in the literature (Li et al. 2015). The equation of motion for a multibody drivetrain model subjected to kinematic joint constraints can then be defined as

$$\left. \begin{aligned} \mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}_q^T \boldsymbol{\lambda} &= \mathbf{Q}_v + \mathbf{Q}_e \\ \mathbf{C}(\mathbf{q}, t) &= \mathbf{0} \end{aligned} \right\} \quad (10)$$

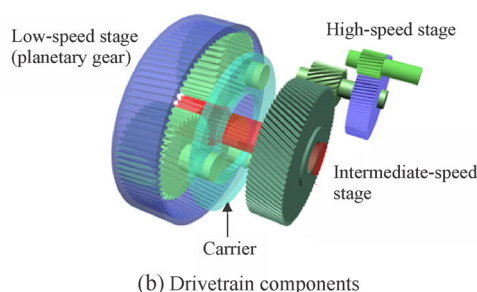
where $\mathbf{C}$ is the vector of the system constraint equations that describe mechanical joints and/or specified motion trajectories, and $\mathbf{q}$ is the vector of the system generalized coordinates. $\mathbf{M}$ is the system mass matrix; $\mathbf{Q}_v$ is the vector of inertia forces that are quadratic in velocity; $\mathbf{Q}_e$ is the vector of the generalized external forces including the gear contact forces. $\mathbf{C}_q$ is the Jacobian matrix of the constraint equations, and $\boldsymbol{\lambda}$ is the vector of Lagrange multipliers used to define the generalized constraint forces.

3.2 Wind turbine drivetrain model and numerical results

To demonstrate the use of the gear dynamics simulation capability discussed in this section, a 750 kW wind turbine drivetrain model utilized in the Gearbox Reliability Collaborative (GRC) project led by NREL is developed (Sheng 2012; Guo et al. 2012a). The drivetrain consists of one planetary gear at the low-speed stage and two parallel axis gears at the intermediate and high-speed stages as shown in Fig. 3 and Table 1. There are three planet gears framed on the carrier, which carries the input load from the wind turbine rotors. The ring gear is fixed to the gear train housing. The sun gear is connected to the input shaft of the intermediate-speed stage parallel axis gear. The overall gear ratio is 81.49. The gear geometry parameters are summarized in Table 2 (Sheng 2012; Guo et al. 2012a). The mass and



(a) NREL drivetrain dynamometer test (photo from (Guo et al. 2012a))



(b) Drivetrain components

Fig. 3 750 kW wind turbine drivetrain

Table 1 750 kW wind turbine specification (Sheng 2012)

Type	Three blade up wind
Power rating	750 KW
Rotor diameter	48.2 m
Rated rotor speed	22 rpm
Power regulation	Stall
Tower	Welded tubular steel
Nominal hub height	55 m
Blade length	23.5 m
Gearbox ratio	1:81.491
Rated wind speed	16 m/s
Service life	20 years

inertia properties of the gears as well as the bearing support stiffness are found in literature (Guo et al. 2012a).

To discuss accuracy of helical gear tooth contact model, the tooth contact force of the sun gear that is engaged with a planet gear is compared with the finite element model using ABAQUSTM. A constant torque of 5122 Nm is applied to evaluate the tooth contact force for one mesh cycle as shown in Fig. 4. It is observed from this figure that the tooth contact force gradually increases to the maximum value and then decreases due to the gradual tooth engagement of helical gears. The contact forces are shared with the adjacent tooth in the result at -16 and $+14$ degree rotation angles as shown in Fig. 4. On the other hand, only one tooth is in contact at 0 degree, resulting in the maximum tooth contact force. Furthermore, using the sliced gear tooth surface, the contact force distribution across the helical gear face width is also well captured as shown in Fig. 4. Since the gear tooth surface is cut into multiple sliced profiles across the width, each sliced profile comes into contact one by one during a mesh cycle and it causes impulsive forces as exhibited in the resultant contact forces obtained using the dynamics simulation in Fig. 4. This effect can be decreased by further increasing the number of slices.

Furthermore, to validate the drivetrain model, measured planet gear bearing load data of the NREL dynamometer test (Guo et al. 2012b) are compared with those of the drivetrain dynamic simulation. The measured input torque is approximated in this study by a sinusoidal function as shown in Fig. 5a and the bearing load in the front side (called upwind bearing load) is compared in Fig. 5b, showing good agreement in magnitude and frequency with the test results.

4 Probabilistic contact fatigue damage

4.1 Contact fatigue model

Using the wide spatiotemporal wind uncertainty model introduced in Section 2, a numerical procedure for predicting the

Table 2 Wind turbine drivetrain specification (Sheng 2012; Guo et al. 2012a)

Components		Tooth number	Normal module	Root diameter	Pressure angle	Helix angle	Face width
Low-speed stage	Ring gear	99	10	1047 mm	20°	7.5° L	230 mm
	Planet gear	39	10	372 mm	20°	7.5° L	227.5 mm
	Sun gear	21	10	186 mm	20°	7.5° R	220 mm
Intermediate-speed stage	Gear	82	8.25	678 mm	20°	14° R	170 mm
	Pinion	23	8.25	174 mm	20°	14° L	186 mm
High-speed stage	Gear	88	5	440 mm	20°	14° L	110 mm
	Pinion	22	5	100 mm	20°	14° R	120 mm

gear tooth contact fatigue life is discussed in this section. The contact fatigue is a typical failure mode exhibited in properly lubricated gears and is classified as the subsurface-initiated failure (Choi and Liu 2006). That is, the total fatigue life is defined by the sum of the number of load cycles required to

initiate the subsurface crack N_i and that required for the crack to propagate to the surface N_p as (Osman and Velez 2011; Choi and Liu 2006)

$$N = N_i + N_p \quad (11)$$

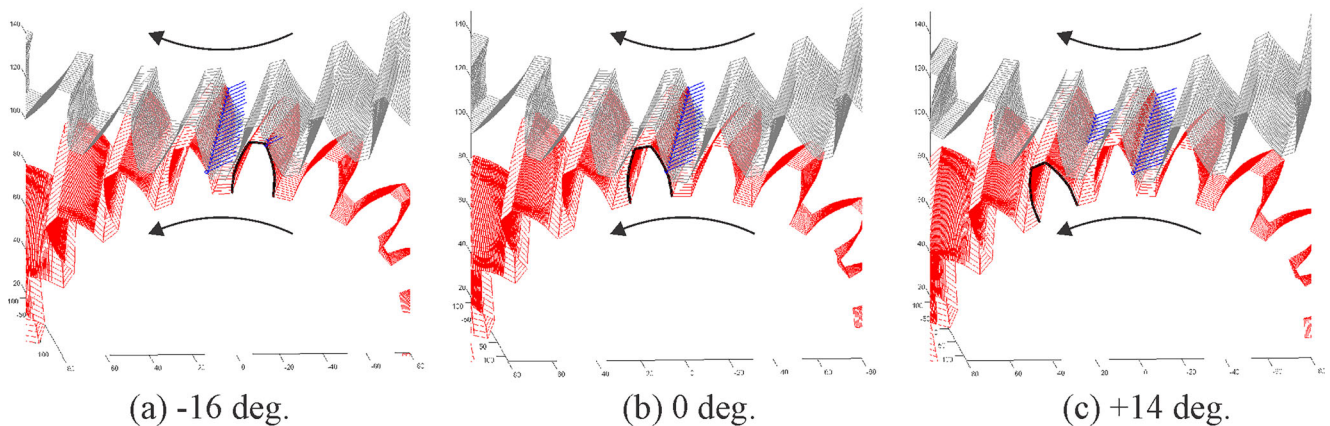
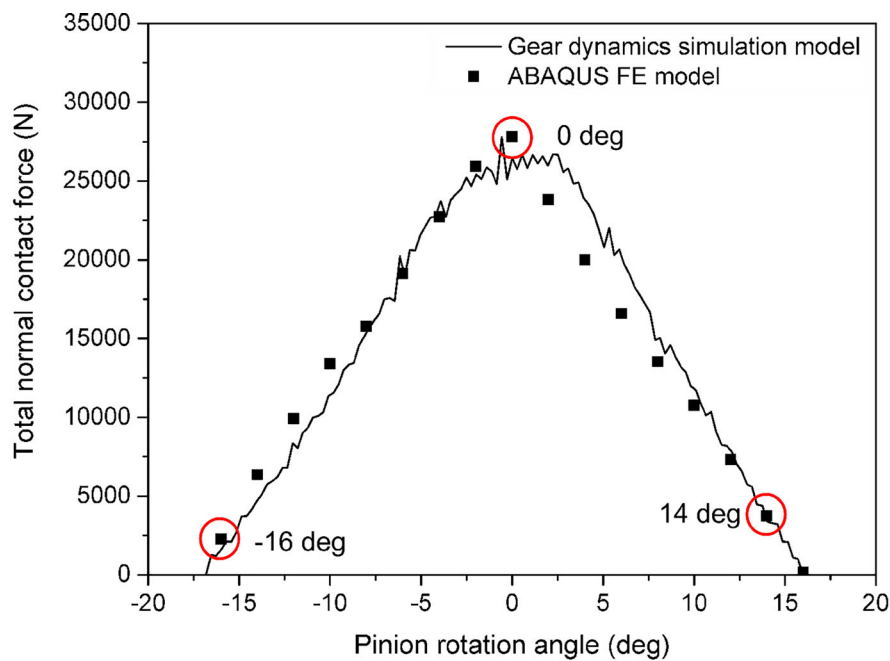
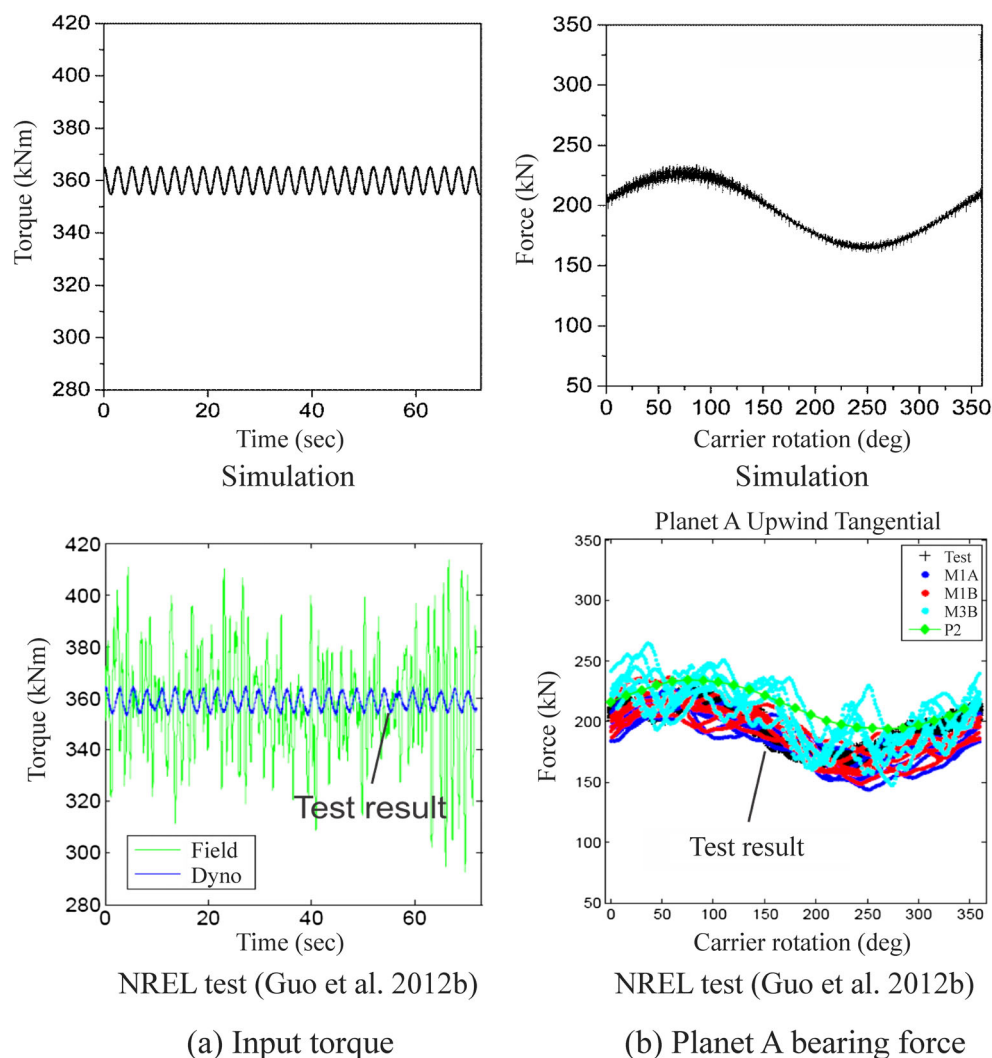
**Fig. 4** Contact force variation for a mesh cycle and comparison with finite element model

Fig. 5 Planetary gear bearing load comparison (NREL test results from (Guo et al. 2012b))



Using the Dang Van's assumption that the fatigue micro-crack appears when the mono-crystal reaches the elastic shakedown due to shearing, the number of cycles for crack initiation is defined by (Osman and Velez 2011):

$$N_i = \frac{1}{2} \left(\frac{\tau_{\max} + 3(\tau'_f/\sigma'_f - 0.5)p_H}{\tau'_f} \right)^{1/c} \quad (12)$$

where $\tau_{\max}$ and p_H are, respectively, the maximum shear stress and hydrostatic stress in the subsurface; σ'_f and τ'_f are fatigue strength coefficients for tension/compression and shearing, respectively; and c is a fatigue strength exponent. It is assumed that the crack is initiated at a point where the ratio of the maximum shear stress to the hardness is maximal and the initial crack is parallel to the surface (Choi and Liu 2006; Jiang et al. 1993) as shown in Fig. 6. Assuming the Hertzian contact between two surfaces, the maximum contact pressure $p_{\max}$ can be obtained as a function of the normal contact force F_N predicted

by the gear dynamics simulation. The greatest maximum shear stress beneath the surface is then obtained as $\tau_{\max} = 0.3p_{\max}$ at the $z_0 = 0.786 b_0$ from the surface, where b_0 is the half width of the contact patch (Budynas and Nisbett 2008).

On the other hand, the number of cycles that causes the initial crack, parallel to the surface, to reach the surface is calculated using the Paris equation by (Osman and Velez 2011; Dong et al. 2013)

$$N_p = \int_{a_0}^{a_c} \frac{1}{C_p((\Delta K)^m - (\Delta K_0)^m)} da_p \quad (13)$$

where a_0 is the half length of the initial crack and a_c is half of the critical crack length given by $a_c = z_0/\sin \alpha$ as shown in Fig. 6 (Osman and Velez 2011; Dong et al. 2013). In the preceding equation, ΔK is the model II stress intensity factor that is quadratic in the ratio of maximum shear stress to the hardness, while ΔK_0 is the threshold for the crack growth (Osman and Velez 2011; Choi and Liu 2006). C_p and m are

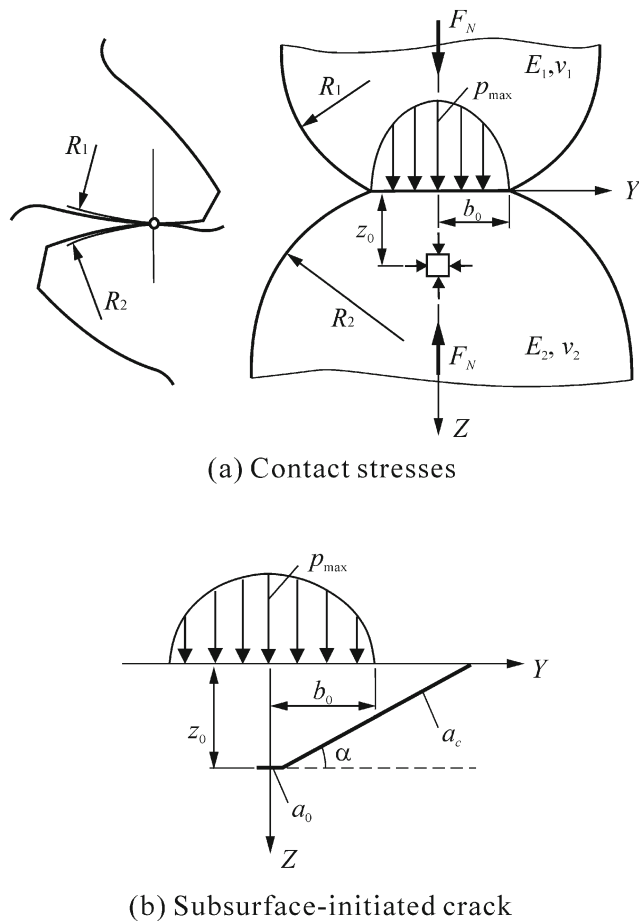


Fig. 6 Contact stresses and crack and subsurface-initiated crack

constants that are identified experimentally (Osman and Velez 2011; Kato et al. 1993).

4.2 Probability of contact fatigue failure under wind load uncertainty

Using the 10-min gear dynamics simulation under a given wind load scenario characterized by v_{10}^i and i_{10}^j , the contact fatigue damage for all teeth of a gear with a design variable $\mathbf{d}$ is evaluated by Miner's rule (Melchers 1999) as

$$D_{10\min}^{ij}(\mathbf{d}, v_{10}^i, i_{10}^j) = \sum_{k=1}^{n_{10}^{ij}} \frac{1}{N^{ijk} \left(\frac{p_{\max}^{ijk}}{p_{\max}} \right)} \quad (14)$$

where N^{ijk} is the number of load cycles to failure for each meshing cycle k and n_{10}^{ij} is the number of load cycles of the gear tooth under consideration in the 10-min simulation. That is, N^{ijk} is defined using the maximum contact pressure $p_{\max}^{ijk}$ at each load cycle k under the 10-min wind scenario given by v_{10}^i and i_{10}^j . The fatigue damage is evaluated at several points on the gear tooth such as the engage, pitch and recess points (Osman and Velez 2011; Dong et al. 2013). The critical point

on the critical tooth exhibiting the largest damage is selected and used for the constraint evaluation in the RBDO process. Using the preceding equation, the 10-min contact fatigue damage in the l -th year can be defined as follows:

$$D_{10\min}^l(\mathbf{d}; \mathbf{y}^l) = \int_{I_L}^{I_U} \int_{V_L}^{V_U} f_{VI}^{ijl}(v_{10}^i, i_{10}^j; \mathbf{y}^l) \cdot D_{10\min}^{ij}(\mathbf{d}, v_{10}^i, i_{10}^j) dv_{10} di_{10} \quad (15)$$

where V_L and V_U indicate the lower and upper bounds of the mean wind speed under consideration, while I_L and I_U are those of the turbulence intensity. $f_{VI}^{ijl}(v_{10}^i, i_{10}^j; \mathbf{y}^l)$ is the joint PDF of an annual wind load for $\mathbf{y}^l = [C^l \ k^l \ a^l \ b^l \ \tau^l]^T$ in the l -th year. By numerically integrating the preceding equation using Riemann integral, one-year fatigue damage in the l -th year is obtained as

$$D_{1\text{year}}^l(\mathbf{d}; \mathbf{y}^l) = 6 \times 3000 \sum_{i=1}^{n_v} \sum_{j=1}^{n_i} f_{VI}^{ijl}(v_{10}^i, i_{10}^j; \mathbf{y}^l) \cdot D_{10\min}^{ij}(\mathbf{d}, v_{10}^i, i_{10}^j) \Delta v_{10} \Delta i_{10} \quad (16)$$

where it is assumed that a wind turbine is operated for 3000 h per year (Lesmerises and Crowley 2013), considering the fact that the gear teeth do not experience loading during the maintenance period and when the wind speed is lower than the cut-in wind speed or higher than the cut-out speed. To evaluate the one-year fatigue damage for given design variable $\mathbf{d}$, the 10-min damage $D_{10\min}^{ij}(\mathbf{d}, v_{10}^i, i_{10}^j)$ needs to be calculated for $n_v \times n_i$ sampling points for v_{10}^i ($i = 1, \dots, n_v$) and i_{10}^j ($j = 1, \dots, n_i$) using the 10-min drivetrain dynamics simulation.

To account for wind load variability over 20 years (i.e., target service life of the wind turbine system), randomness of the joint PDF parameters of wind loads $\mathbf{y}$ over 20 years are defined as

$$\mathbf{Y} = [C^T \ k^T \ a^T \ b^T \ \tau^T]^T \quad (17)$$

where the preceding vector contains 20 random variables for each parameter as $\mathbf{C} = [C^1 \ C^2 \ \dots \ C^{20}]^T$, $\mathbf{k} = [k^1 \ k^2 \ \dots \ k^{20}]^T$, $\mathbf{a} = [a^1 \ a^2 \ \dots \ a^{20}]^T$, $\mathbf{b} = [b^1 \ b^2 \ \dots \ b^{20}]^T$, and $\mathbf{\tau} = [\tau^1 \ \tau^2 \ \dots \ \tau^{20}]^T$ (Hu et al. 2016a). Accordingly, the 20-year gear contact fatigue damage considering wind load uncertainty can then be defined as

$$D_{20\text{year}}(\mathbf{d}, \mathbf{Y}) = \sum_{l=1}^{20} D_{1\text{year}}^l(\mathbf{d}; \mathbf{y}^l) \quad (18)$$

Using the preceding equation, the probability of gear contact fatigue failure can be evaluated using Monte Carlo simulation (MCS) as follows (Hu et al. 2016a):

$$P(D_{20\text{year}}(\mathbf{d}; \mathbf{Y}) > 1) \cong \frac{1}{\text{NMCS}} \sum_{i=1}^{\text{NMCS}} I_{\Omega_F}(\mathbf{d}^i, \mathbf{y}^i) \quad (19)$$

where $\mathbf{d}^i$ and $\mathbf{y}^i$ are, respectively, the i -th realization of $\mathbf{d}$ and $\mathbf{Y}$ selected randomly from the corresponding PDFs; NMCS is the number of realizations for MCS; and Ω_F is the failure domain defined by $D_{20\text{year}}(\mathbf{d}; \mathbf{Y}) > 1$; and I_{Ω_F} is an indicator function defined as follows:

$$I_{\Omega_F}(\mathbf{d}^i, \mathbf{y}^i) = \begin{cases} 1 & \text{for } (\mathbf{d}^i, \mathbf{y}^i) \in \Omega_F \\ 0 & \text{otherwise} \end{cases} \quad (20)$$

It is important to notice here that $D_{20\text{year}} > 1$ indicates that fatigue life is less than 20 years.

5 RBDO of wind turbine drivetrain

5.1 Design variables

In this section, RBDO of a wind turbine drivetrain is discussed such that 20-year contact fatigue life can be assured with the target reliability while minimizing the total weight of the drivetrain. To this end, the face width and amount of the tip relief (profile modification) are selected as design variables. To demonstrate the effect of these two design variables on the contact fatigue life, the maximum contact pressure on a tooth surface of the sun gear at the planetary stage for one mesh cycle is shown in Fig. 7 for four different face widths without tip relief. It is important to notice here that the sun gear is engaged with three planet gears and its root diameter is smallest at the planetary stage, thus it is the most critical gear for contact fatigue. In this figure, P_1 is the engage point, P_2 is the pitch point, and P_3 is the recess point (Dong et al. 2013). In the baseline design, the face width is 220 mm with no tip relief (i.e., involute profile) (Sheng 2012). The rated mean wind

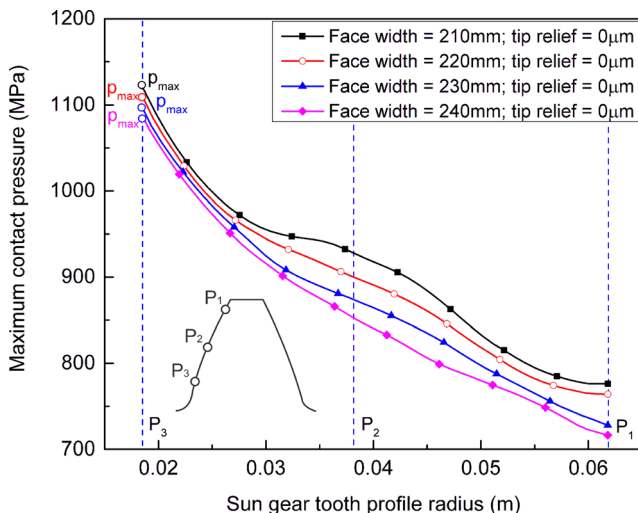


Fig. 7 Effect of face width on maximum contact pressure

speed of 16 m/s and turbulence intensity of 0.14 categorized in the medium range based on an ISO standard (International Organization for Standardization 2005) are assumed in this example. It is observed from this figure that the recess point P_3 has the maximum contact pressure regardless of the face width. Larger face width leads to smaller maximum contact pressure on the tooth surface, resulting in smaller damage as shown in Fig. 8. However, an increase in face width leads to an increase of the gear weight, giving an adverse impact on the objective function. On the other hand, effect of the tip relief on the maximum contact pressure is shown in Fig. 9 for six different tip relief amounts with a face width of 220 mm. The tip relief starts from the pitch point P_2 . It is observed from this figure that an increase of the tip relief reduces the maximum contact pressure at the recess point, while further increase of the tip relief shifts the maximum contact pressure point to the pitch point and then the maximum contact pressure starts increasing from the tip relief amount of 60 μm . Therefore, there is an optimum tip relief amount that can minimize the greatest maximum contact pressure as shown in Fig. 10 and an optimum tip relief needs to be determined considering wind load uncertainty. Furthermore, since the material removal for the tip relief is on the order of microns, the tip relief has no effect on the total weight (i.e., the objective function).

5.2 Formulation of wind turbine drivetrain RBDO

Using the integrated numerical procedure for evaluating the gear tooth contact damage under wide spatiotemporal wind load uncertainty presented in the preceding sections and the design variables selected, the following RBDO problem is formulated:

$$\begin{aligned} &\text{Minimize } J(\mu_B) \\ &\text{Subject to } P(D_{20\text{year}}(\mathbf{d}; \mathbf{Y}) > 1) \leq P_F^{\text{Tar}} \\ &\quad \text{for } \mu_B^L \leq \mu_B \leq \mu_B^U \text{ and } \mu_{\delta_a}^L \leq \mu_{\delta_a} \leq \mu_{\delta_a}^U \end{aligned} \quad (21)$$

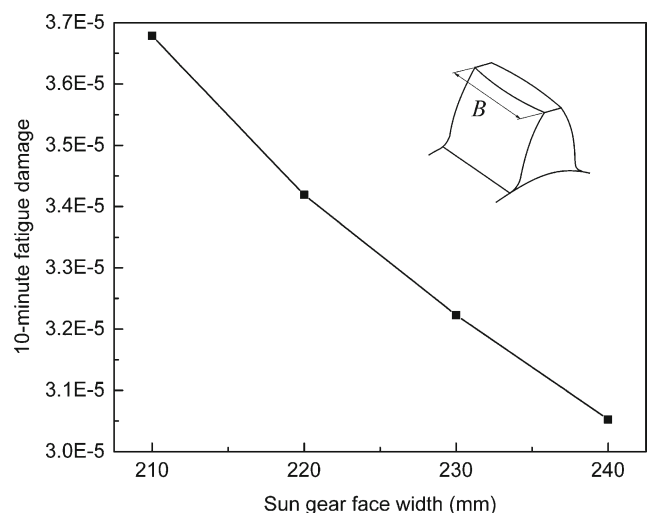


Fig. 8 Effect of face width on 10-min fatigue damage

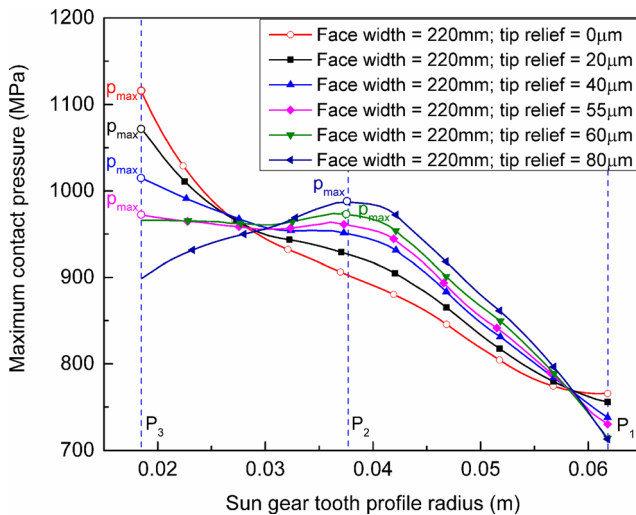


Fig. 9 Effect of tip relief on maximum contact pressure

where P_F^{Tar} is the target probability of failure; μ_B and μ_{δ_a} are means of the random face width B and the random tip relief amount δ_a of the sun gear, respectively; the lower and upper bounds of the design variables are assumed to be $167.5 \leq \mu_B \leq 272.5$ mm and $0 \leq \mu_{\delta_a} \leq 100$ μm , respectively; and the cost function $J(\mu_B)$ is defined by the total mass of the planetary gear consisting of a ring gear, a sun gear and three planet gears as

$$J(\mu_B) = m_{\text{ring}}(\mu_B) + m_{\text{sun}}(\mu_B) + 3m_{\text{planet}}(\mu_B) \quad (22)$$

In the RBDO problem shown in Eq. (21), uncertainty of the design variables (i.e., the face width and tip relief amount) is modeled by Gaussian distribution as shown in Table 3. Since small gear tooth manufacturing variance can lead to unintended change in the contact force distribution over the contact surface, gear tooth geometric variance may also need to be considered. Such a geometric uncertainty associated with manufacturing tolerance, which is not considered in this study,

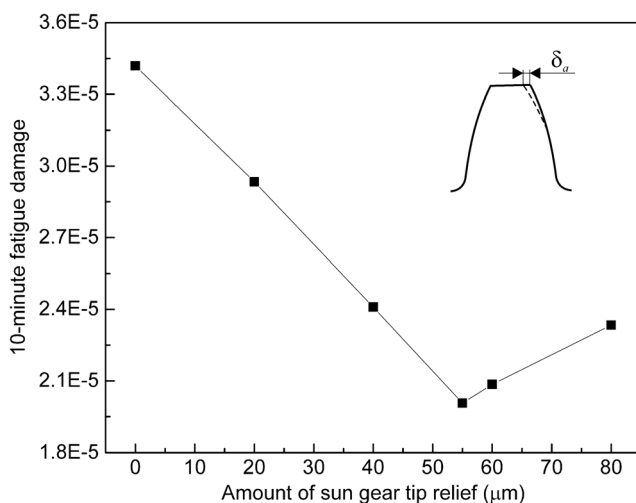


Fig. 10 Effect of tip relief on 10-min fatigue damage

Table 3 Standard deviations and tolerances of gear design variables

	Standard deviation	Tolerance
Face width (mm)	1.2	3.6
Tip relief (μm)	1.5	4.5

is defined as a random field and various stochastic field models have been proposed in the literature (Lazarov et al. 2012; Zhang and Kang 2016).

Wind turbine drivetrain RBDO in Eq. (21) is carried out using sampling-based RBDO method with Sequential Quadratic Programming (SQP) algorithm in RAMDO software (Reliability and Multidisciplinary Design Optimization (RAMDO) Software 2016). The sampling-based RBDO method calculates the probability of failure $P(D_{20\text{year}}(\mathbf{d}; \mathbf{Y}) > 1)$ using MCS as indicated in Eq. (19), and estimates stochastic design sensitivity of the probability of failure using score function method (Lee et al. 2011). Accuracy of the calculated probability of failure and the stochastic design sensitivity depends on the number of MCS samples, NMCS. It is important to notice here that constraint functions need to be evaluated NMCS times, thus use of the large number of MCS samples causes a computational burden, making the RBDO process very computationally intensive. In particular, for the wind turbine drivetrain considered in this study, the contact fatigue damage needs to be evaluated at each design of experiment (DOE) point using the nonlinear multibody dynamics simulation for various 10-min wind load scenarios to account for wind load uncertainty as well as design variable uncertainty. To reduce the computational cost, surrogate models for evaluating the 20-year contact fatigue damage are generated and utilized in this study for the sampling-based RBDO.

5.3 Surrogate model for contact fatigue damage calculation

There are a wide variety of surrogate models proposed in the past such as polynomial response surfaces (Faravelli 1989), support vector regression (SVR) and classifiers (SVC) (Smola and Schölkopf 2004), neural networks (Hurtado and Alvarez 2001), radial basis function (Park and Sandberg 1993), and Kriging predictors (Janusevskis and Riche 2013). The Kriging method has been widely used due to its capability of dealing with nonlinear problems (Zhao et al. 2011). A Kriging method constructs the mean structure using a fixed set of polynomial basis functions. The dynamic Kriging method proposed in the literature (Zhao et al. 2011), on the other hand, obtains an optimum mean structure using basis functions selected by a genetic algorithm, and an accurate optimum of the correlation parameters are obtained through the generalized pattern search algorithm, leading to better accuracy for highly nonlinear problems (Sen et al. 2015). Thus, in this study, the dynamic

Kriging method in the RAMDO software is utilized to generate a surrogate model for prediction of the wind turbine gear contact fatigue damage.

For efficient RBDO process, initial design of RBDO is set to the deterministic design optimization (DDO) optimum (Youn et al. 2005). While a local surrogate model is often generated and used at the each design point within the target reliability range of random design variables (Hu et al. 2016a), a surrogate model covering relatively wider range of design variables – mid-size window – around the DDO optimum design is generated in this study to reduce the computation cost associated with the surrogate model generation. That is, surrogate models are generated only once prior to the RBDO process and then used throughout the RBDO iteration process. To ensure its accuracy, sampling points are selected using Latin Centroidal Voronoi Tessellations (LCVT) sampling method with hypercube window. The window covers the twelve sigma (12σ) range, meaning that approximately one ninth of the entire design variable domain can be covered. The center of the window is intentionally shifted from the DDO optimum by three sigma of the face width. This is attributed to the fact that the gear face width tends to increase from the DDO optimum point to meet the higher target reliability requirement in RBDO. The window has to be properly selected such that the design variables taken in the RBDO iterative process can lie inside the window to assure the accuracy. The window size has also to be small enough to minimize the number of DOE points to reduce the computational cost for the surrogate model generation. Since surrogate models are generated only once, this leads to a significant reduction of CPU time as compared to that of local surrogate models. While the number DOE points in 12σ window must be larger than that of the local surrogate model to ensure the accuracy, simulations at all DOE points can be efficiently run in parallel using high-performance computing (HPC) techniques.

In this study, 50 DOE points are selected using LCVT sampling method with hypercube window in the 12σ range. The wind turbine drivetrain dynamics simulation is carried out at the 50 DOE points to obtain 50 fatigue damage response vectors. To account for the wind load uncertainty, each response vector contains 10-min gear contact fatigue damages for all wind load cases characterized by v_{10}^i ($i = 1, \dots, n_v$) and i_{10}^j ($j = 1, \dots, n_i$). For each wind load scenario (i.e., a combination of v_{10}^i and i_{10}^j), one surrogate model is generated to predict the corresponding 10-min contact fatigue damage for the given design point. Using the 10-min fatigue damage response vectors generated, $n_v \times n_i$ surrogate models associated with $n_v \times n_i$ wind load scenarios are generated by the dynamic Kriging method in RAMDO software. That is, the surrogate model involves the random 10-min time-domain wind speed data generation by TurbSim, the 10-min time-domain coupled nonlinear aero-hydro-servo-elastic simulation of a wind turbine by FAST, the 10-min time-domain multibody

drivetrain dynamics simulation discussed in Section 3, and the 10-min fatigue damage calculation discussed in Section 4 as summarized in Fig. 11.

5.4 RBDO numerical procedure

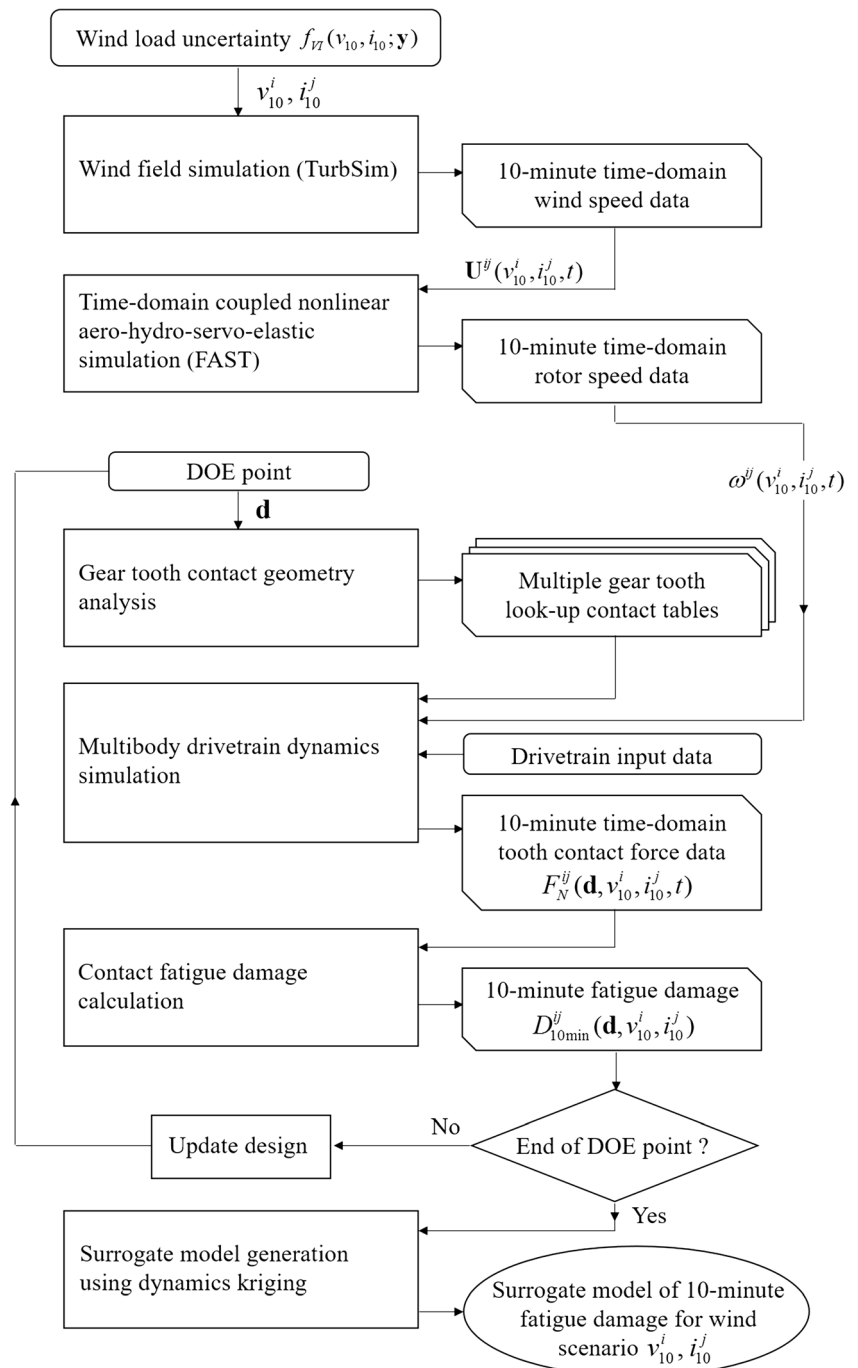
A numerical procedure of the wind turbine drivetrain RBDO using the mid-size surrogate model is presented in Fig. 12 and summarized as follows:

- Step 1: Using the DDO optimum obtained by the averaged wind load uncertainty model as summarized in Appendix (Li et al. 2016), $n_v \times n_i$ surrogate models for the 10-min fatigue damage using 50 DOE points are generated for wind load scenarios characterized by v_{10}^i ($i = 1, \dots, n_v$) and i_{10}^j ($j = 1, \dots, n_i$) using the dynamic Kriging method.
- Step 2: The RBDO process starts from the DDO optimum. At every design, NMCS realizations of design variables (i.e., $\mathbf{d}^i$ for $i = 1, \dots, \text{NMCS}$) are created. In this study, $\text{NMCS} = 500,000$ and Gaussian distribution are assumed for modeling variability of design variables as shown in Table 3 for the face width and tip relief amount. The NMCS realizations are inputted to the surrogate models to obtain the NMCS 10-min fatigue damage tables, each of which contains $n_v \times n_i$ 10-min fatigue damages under $n_v \times n_i$ wind load conditions.
- Step 3: NMCS realizations of 20 sets of the joint PDF parameters $\mathbf{Y} = [\mathbf{C}^T \quad \mathbf{k}^T \quad \mathbf{a}^T \quad \mathbf{b}^T \quad \boldsymbol{\tau}^T]^T$ are created and then $\text{NMCS} \times 20$ wind load probability tables are generated, each of which contains the probabilities of $n_v \times n_i$ wind load conditions.
- Step 4: NMCS 20-year fatigue damages are calculated, and then the probability of failure $P(D_{20\text{year}}(\mathbf{d}; \mathbf{Y}) > 1)$ and its stochastic design sensitivity are evaluated at the current design. If the convergence is achieved, the RBDO iteration stops; otherwise, the design variables are updated and then the RBDO process is repeated from Step 2 to 4 until the RBDO optimum design is achieved.

6 RBDO results and discussion

For the 750 kW wind turbine drivetrain described in Section 3, RBDO of the sun gear at the planetary stage is carried out. To define wind load probability tables, the mean wind speed range is defined from 5 m/s to 25 m/s with an increment of 2 m/s, while the turbulence intensity range is defined from 0.02 to 1 with 0.02 increment. In order to illustrate the joint PDF of wind load $f_{VI}^{ijl}(v_{10}^i, i_{10}^j; \mathbf{y}^l)$ for different joint PDF parameters $\mathbf{y} = [C \quad k \quad a \quad b \quad \tau]^T$, joint PDFs of two different cases

Fig. 11 Flowchart of surrogate model generation process for a wind scenario characterized by v_{10}^i and i_{10}^j

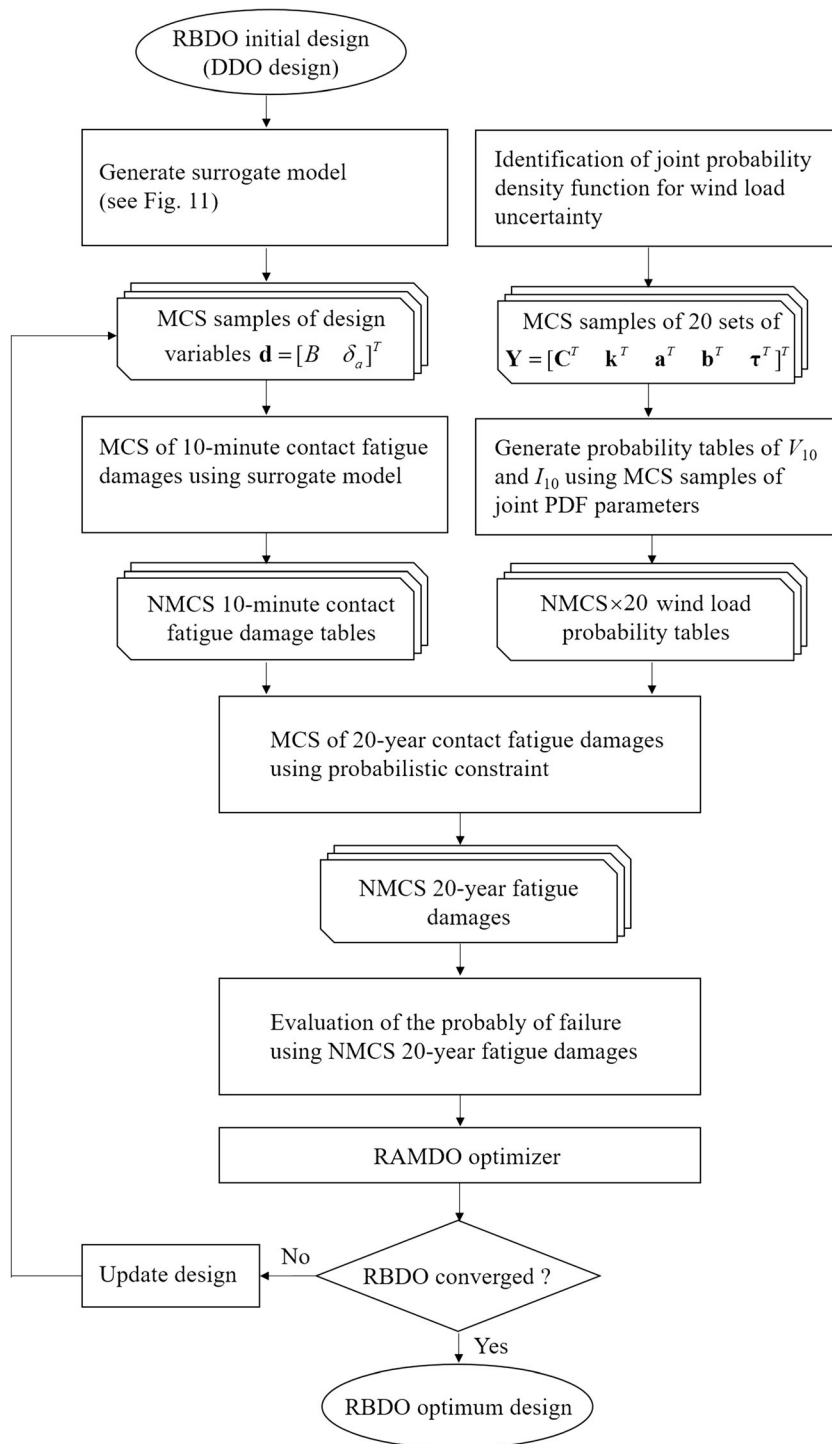


(called PDF1 and PDF2) are presented in Figs. 13a and 14a, respectively. In PDF1, the joint PDF parameters are given as $\mathbf{y}_{PDF1} = [6.00 \ 1.75 \ 2.00 \ 0.20 \ 0.30]^T$ and high probability region lies in the low wind speed range ($5 \text{ m/s} < V_{10} < 10 \text{ m/s}$) and in the low turbulence intensity range ($0.0 < I_{10} < 0.1$). The highest probability density of 1.22 occurs at $V_{10} = 5 \text{ m/s}$ and $I_{10} = 0.04$, where both mean wind speed and turbulence intensity are low. On the other hand, the joint PDF parameters of PDF2 are given as $\mathbf{y}_{PDF2} = [10.00 \ 2.75 \ 4.00 \ 0.40 \ 0.70]^T$ and high probability

region lies in the mild wind speed range ($7 \text{ m/s} < V_{10} < 17 \text{ m/s}$) and in the mild turbulence intensity range ($0.1 < I_{10} < 0.3$). The highest probability density of 1.21 occurs at $V_{10} = 13 \text{ m/s}$ and $I_{10} = 0.2$. The mean wind speed and turbulence intensity are higher than those of PDF1.

Since one-year contact fatigue damage in the l -th year is evaluated by the integral of the product of the 10-min damage and the joint PDF of wind loads (i.e., $f_{V_l}^{ijl}(v_{10}^i, i_{10}^j; \mathbf{y}^l) \cdot D_{10min}^{ij}(d, v_{10}^i, i_{10}^j)$) over the mean wind speed and turbulence intensity ranges, this product is shown for these two wind load

Fig. 12 Flowchart of RBDO process



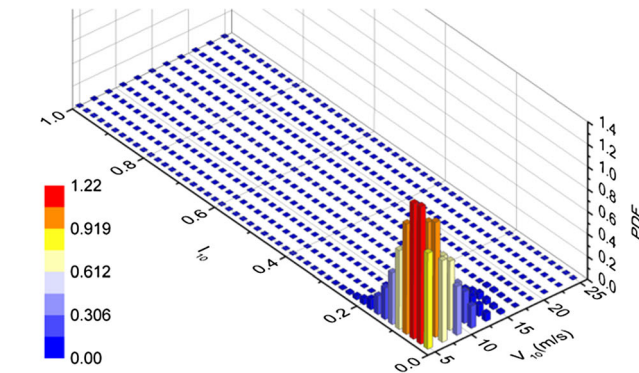
cases in Figs. 13c and 14c, respectively. In this figure, the 10-min fatigue damages $D_{10\min}^{ij}(\mathbf{d}, v_{10}^i, i_{10}^j)$ at the recess point on the most critical tooth of the sun gear are used and plotted in Fig. 13b and 14b. It is observed from this figure that the fatigue damage increases with an increase of the mean wind speed since the number of load cycles and contact loads increase as the rotor speed increases. However, the fatigue damage plateaus when the wind speed gets higher than the rated

speed of 16 m/s. This is attributed to the fact that the blade pitch control is activated at the rated speed (16 m/s) to maintain constant power generation at a constant rotor speed (Muljadi and Butterfield 2000). It is observed from Figs. 13c and 14c that the product of the 10-min damage and the joint PDF of wind loads differ significantly due to the difference in the wind load variability distribution, thereby resulting in noticeably different one-year contact fatigue damages as $D'_{1\text{year}}$

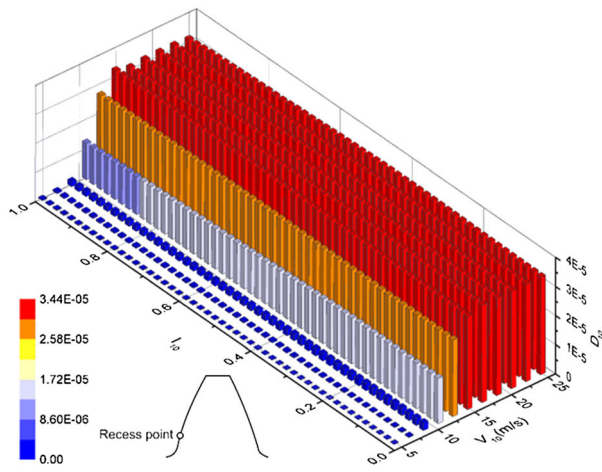
$(\mathbf{d}; \mathbf{y}_{\text{PDF1}}) = 3.65 \times 10^{-2}$ and $D_{1\text{year}}^I(\mathbf{d}; \mathbf{y}_{\text{PDF2}}) = 1.59 \times 10^{-1}$ for the sun gear. That is, consideration of wide spatio-temporal wind load distributions is crucial to realistic prediction of the gear tooth contact fatigue life for RBDO of wind turbine drivetrains.

To find the initial value for RBDO process, DDO is carried out. In DDO, the averaged wind load probability model

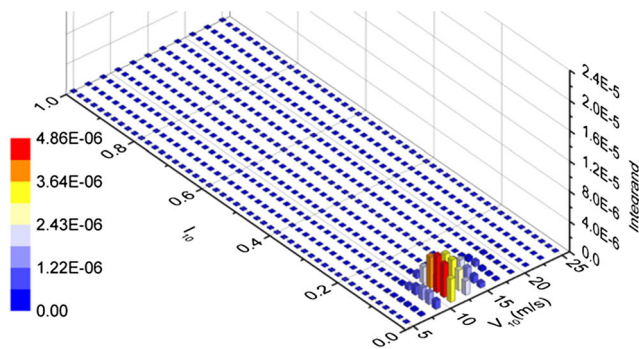
developed using Monte Carlo simulation of the joint PDFs is utilized to calculate the 10-min contact fatigue damage (Li et al. 2016). Accordingly, the wind load variability over 20 years as well as design variable variability are not considered in DDO for predicting the contact fatigue damage constraint. The formulation and numerical procedure of DDO are summarized in Appendix (Li et al. 2016). This resulted in an



(a) Wind load joint PDF

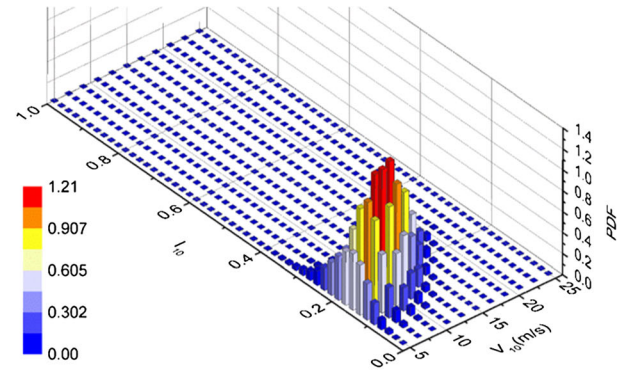


(b) 10-minute contact fatigue damage

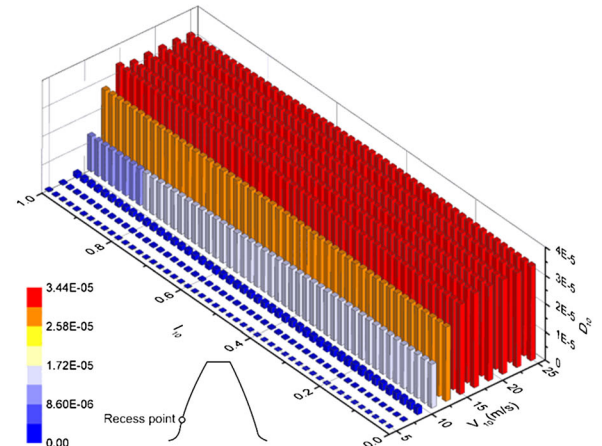


(c) Product of joint PDF and 10-min fatigue damage

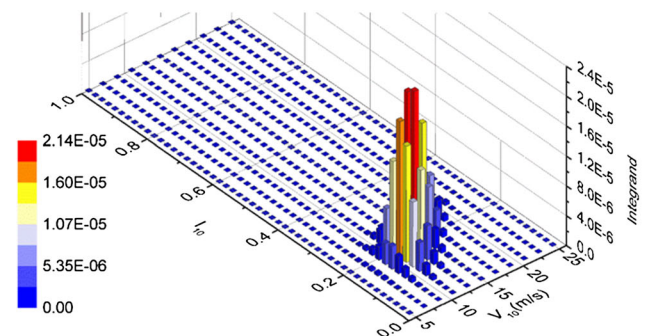
Fig. 13 PDF of the wind load, 10-min damage and their product for $\mathbf{y}_{\text{PDF1}} = [6.00 \ 1.75 \ 2.00 \ 0.20 \ 0.30]^T$



(a) Wind load joint PDF



(b) 10-minute contact fatigue damage



(c) Product of joint PDF and 10-min fatigue damage

Fig. 14 PDF of the wind load, 10-min damage and their product for $\mathbf{y}_{\text{PDF2}} = [10.00 \ 2.75 \ 4.00 \ 0.40 \ 0.70]^T$

optimum face width of 202 mm and an optimum tip relief of 41 μm from the baseline design of 220-mm face width with no tip relief. It is important to notice here that the face width of the sun gear needed to be increased to 231 mm if only face width is selected as the design variable, leading to 4.7% increase in weight. On the other hand, the face width can be decreased to 202 mm by introducing the tip relief to meet the contact fatigue constraint. Since the optimum tip relief allowed for minimizing the greatest maximum shear stress without relying heavily on face width widening to meet the 20-year fatigue life constraint, it led to the weight reduction by 8.4%. With the DDO optimum obtained, the reliability analysis was performed considering the wide spatiotemporal wind load uncertainty over 20 years and the gear design variable uncertainty using the surrogate models, for which 500,000 MCS realizations of design points are created based on the joint PDF of the face width B and the tip relief amount δ_a at each design iteration to calculate the probability of failure. The standard deviations of B and δ_a are assumed as previously presented in Table 3. The probability of failure of the DDO optimum turned out to be 47.7%.

To improve the reliability of the DDO design, the RBDO optimum is sought with the target probability of failure of $P_F^{\text{Tar}} = 2.275\%$ (two sigma quality level) with 500,000 MCS samples. The RBDO result obtained is presented in Table 4 and compared to the baseline design as well as the DDO optimum design. It is observed from this table that the face width and tip relief amount increased from the DDO design to meet the target probability of failure. This led to an increase in weight. However, as compared to the baseline design exhibiting 91.7% probability of failure (i.e., 8.3% reliability), the weight was increased by only 1.4%, while the probability of failure was improved significantly from 91.7% to 2.275% (i.e., 97.725% reliability). This result also indicates the importance of incorporating the tip relief as a design variable and justifies the need for the efficient gear dynamics simulation capability that accounts for the micro-geometry of gear tooth contact. Furthermore, to justify the surrogate model accuracy constructed with the twelve sigma window, the reliability analysis was carried out again at the RBDO optimum point using a local surrogate model constructed by 50 DOE points in the three sigma range and the result was compared with the one obtained using the mid-size surrogate model used in the

RBDO process. The difference in the probability of failure was 0.23%, which is in an acceptable range in practice.

To discuss how the two design variables converged from the baseline design to the RBDO optimum, the face width and the tip relief in the DDO and RBDO iteration processes are presented in Fig. 15, while the corresponding objective function (weight) and the probability of failure are also presented in Fig. 15 as a function of the iteration step. Since the contact fatigue constraint is not met at the baseline design, the tip relief amount is sharply increased to meet the contact fatigue constraint in the DDO process, while the face width is decreased to minimize the objective function as shown in Fig. 15a. However, the fatigue constraint becomes violated at the DDO step 3, thus the face width is increased while decreasing the tip relief to meet the fatigue constraint and to minimize the objective function simultaneously. This resulted in the DDO optimum of 202 mm face width and 41 μm tip relief. Starting from this value, since the probability of failure of the DDO optimum is 47.7% and does not meet the target probability of failure, both face width and the tip relief increase sharply in the first few iterations to meet the target probability of failure as shown in Fig. 15b. Because of the increase in the face width, the objective function increases as well from the DDO optimum. However, it is important to notice here that increase in the objective function (weight) to achieve the targeted 2.275% probability of failure is only 1.4% from the baseline design.

7 Summary and conclusions

Despite a significant need for improving reliability of wind turbine drivetrains to reduce downtime of wind turbine systems caused by gear failures, RBDO of wind turbine drivetrains has not yet been addressed considering wide spatiotemporal wind load uncertainty and variability of gear design variables. This requires developing a multi-physics computational procedure integrated in RBDO process that enables a systematic prediction of probabilistic gear tooth contact fatigue damages using the high-fidelity nonlinear multibody gear dynamics model under wide spatiotemporal wind load uncertainty. To account for wide spatiotemporal wind load variability, randomness of PDF parameters of the wind load joint PDF characterized by the 10-min mean wind speed and turbulence intensity was considered in this study. The spatial time-domain wind data was then generated using TurbSim software and used for performing the time-domain coupled nonlinear aero-hydro-servo-elastic simulation of a wind turbine considering the pitch control of rotor blades using FAST software. The resulting 10-minute wind speed data of the rotor hub were inputted to the high-fidelity multibody drivetrain dynamics simulation for the contact fatigue evaluation under various wind scenario considered in the wind load

Table 4 DDO and RBDO results

	Baseline design	DDO design	RBDO design
Face width (mm)	220	202	223
Tip relief (μm)	0	41	54
Fatigue life (year)	15	20	20
Probability of failure	91.7%	47.7%	2.275%
Normalized mass	1.000	0.916	1.014

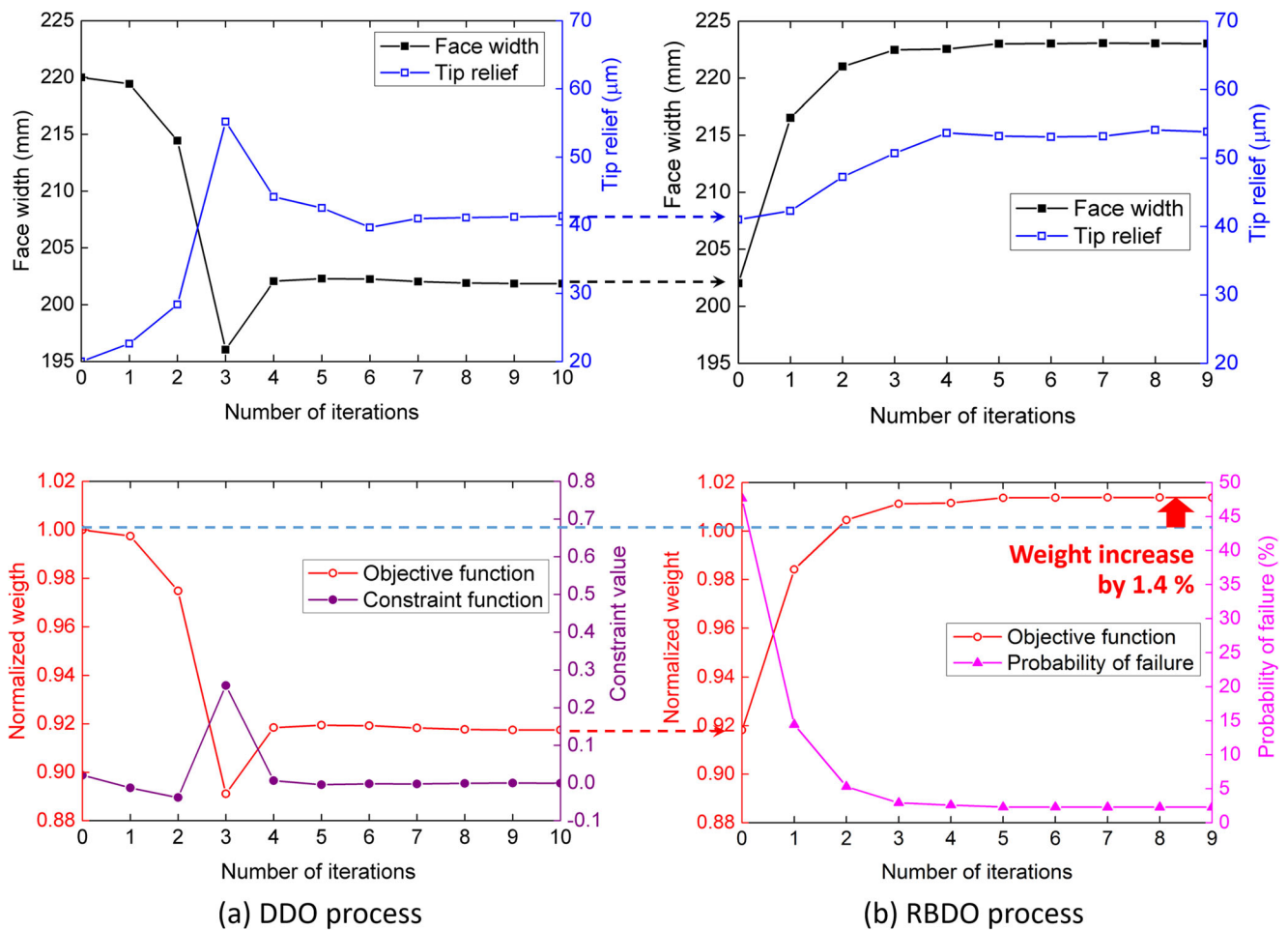


Fig. 15 Design variables, objective function and probability of failure for the DDO and RBDO processes

uncertainty model using MCS. However, this leads to a computationally intensive procedure due to not only the use of MCS, but also the need for considering the precise gear contact geometry in the multibody drivetrain dynamics simulations to incorporate the gear tooth profile optimization. For this reason, a tabular gear tooth contact search algorithm was applied to the modeling of wind turbine drivetrains in this study to improve the computational efficiency while retaining the precise gear tooth contact geometry altered by the tooth profile modification in the order of microns. This enabled the effect of profile modification (i.e., tip relief) on the maximum shear contact stress beneath the contact surface in the probabilistic contact fatigue damage evaluation to be accounted for.

Reliability-based design optimization of a wind turbine drivetrain was formulated such that the probabilistic 20-year gear contact fatigue life can be assured with the target reliability while minimizing the total weight of the drivetrain under wind load uncertainty as well as uncertainty of gear design variables. The face width and amount of the tip relief (profile modification) were selected as design variables. The surrogate models for

evaluating the 20-year contact fatigue damage were generated using the dynamic Kriging method and utilized for sampling-based RBDO. The surrogate model covers the twelve sigma (12σ) range, which corresponds to approximately one ninth of the entire design variable domain. The surrogated model was generated only once prior to the RBDO process. A three-stage drivetrain model of a 750 kW wind turbine was developed and validated against the dynamometer test data and finite element gear model results. To find the initial value for RBDO process, DDO was carried out first using the averaged wind load uncertainty model. That is, the wind load uncertainty over 20 year service life as well as design variable uncertainty were not considered in DDO. Using the DDO optimum tip relief and face width of the sun gear defined as the critical gear at the planetary stage, the weight was reduced by 8.4%. However, the reliability of the DDO design was only 47.7% under the wide spatiotemporal uncertainty as well as the uncertainty of design variables. Using RBDO, on the other hand, the target reliability of 97.725% (2 sigma quality level) was successfully achieved with only

a 1.4% increase in the total weight from the baseline design, which had a reliability of 8.3%. Furthermore, it was shown that introducing the tooth profile (tip relief) as a design variable prevents a large increase of the face width which results in a large increase in the weight of the drivetrain, when satisfying the target reliability against the tooth contact fatigue failure.

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Appendix

In this Appendix, the deterministic design optimization (DDO) process using the averaged wind load uncertainty model is briefly summarized (Li et al. 2016). The averaged wind load probability model developed using Monte Carlo simulation of the joint PDFs for V_{10} and I_{10} is utilized. The averaged wind load probability model can be obtained by one million joint PDFs defined by one millions sets of $\mathbf{y}^l = [C^l \ k^l \ a^l \ b^l \ \tau^l]^T$ ($l = 1, \dots, N_{VI}$) as

$$f_{VI}^{ijl}(v_{10}^i, i_{10}^j; \mathbf{y}^l) = c_{V\Sigma}(v_{10}^i, \sigma_{10}^{ij}; \tau^l) \cdot f_{V10}(v_{10}^i; C^l, k^l) \cdot f_{\Sigma10}(\sigma_{10}^{ij}; a^l, b^l) \cdot v_{10}^i \quad (23)$$

where the PDF is evaluated for v_{10}^i and i_{10}^j . Using the resulting one million joint PDFs, the mean of the PDF evaluated for each v_{10}^i and i_{10}^j are calculated to develop the averaged wind load PDF model used for the design optimization as

$$\bar{f}_{VI}^{ij}(v_{10}^i, i_{10}^j) = \frac{\sum_{l=1}^{N_{VI}} f_{VI}^{ijl}(v_{10}^i, i_{10}^j; \mathbf{y}^l)}{N_{VI}} \quad (24)$$

Using this averaged joint PDF of wind load, the 20 year fatigue damage can be defined from the 10-min damage $D_{10\min}^{ij}$ evaluated for v_{10}^i ($i = 1, \dots, n_v$) and i_{10}^j ($j = 1, \dots, n_i$) as follows:

$$D_{20\text{year}} = 6 \times 3000 \times 20 \sum_{i=1}^{n_v} \sum_{j=1}^{n_i} \bar{f}_{VI}^{ij}(v_{10}^i, i_{10}^j) D_{10\min}^{ij}(\mathbf{d}, v_{10}^i, i_{10}^j) \Delta v_{10} \Delta i_{10} \quad (25)$$

It is important to notice here that the wind load variability over 20 years as well as design variable uncertainty are not considered in the DDO process. Using the preceding probabilistic fatigue damage model with an averaged wind load uncertainty assumption, the deterministic design optimization of

the face width B and tip relief amount δ_a of the sun gear at the planetary stage is formulated such that the total mass of the planetary gear stage can be minimized while satisfying the 20-year service life as follows:

$$\begin{aligned} &\text{Minimize} \quad J(B) \\ &\text{Subject to} \quad D_{20\text{year}}(B, \delta_a) - 1 \leq 0 \\ &\quad \text{for} \quad B^L \leq B \leq B^U \quad \text{and} \quad \delta_a^L \leq \delta_a \leq \delta_a^U \end{aligned} \quad (26)$$

where the cost function defines the total mass of the planetary stage gears as $J(B) = m_{\text{ring}}(B) + m_{\text{sun}}(B) + 3m_{\text{planet}}(B)$. The constraint is imposed to ensure that the contact fatigue life is longer than 20 years (i.e., the 20-year fatigue damage is less than 1). The upper and lower bounds of the face width and tip relief amount are assumed as $167.5 \leq B \leq 272.5$ mm and $0 \leq \delta_a \leq 100$ μm .

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